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RULE · Linear-quadratic system (chiziqli-kvadrat)

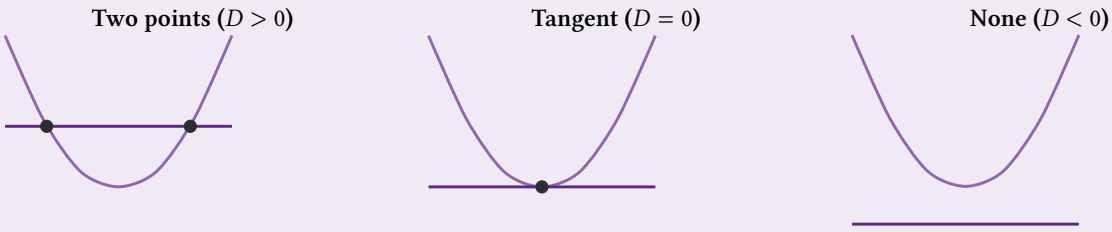
Substitute the linear expression for y into the quadratic. Find the matching y for EACH x.

$y = x + 1,\ y = x^2 - 1 \implies x^2 - x - 2 = 0 \implies x = 2, -1 \implies (2, 3)$ and $(-1, 0)$

RULE · Number of solutions

$D = b^2 - 4ac$ of the resulting quadratic: $D > 0$ two points, $D = 0$ one point (tangent), $D < 0$ none.

$y = 4x - 4,\ y = x^2 \implies x^2 - 4x + 4 = 0,\ D = 0 \implies$ tangent at $(2, 4)$



RULE · Quadratic-quadratic system (kvadrat-kvadrat)

Same leading coefficient → subtract the equations to cancel x^2, leaving a linear equation.

$y = x^2 + 3x + 2,\ y = x^2 - x + 10 \implies 4x = 8 \implies x = 2 \implies (2, 12)$